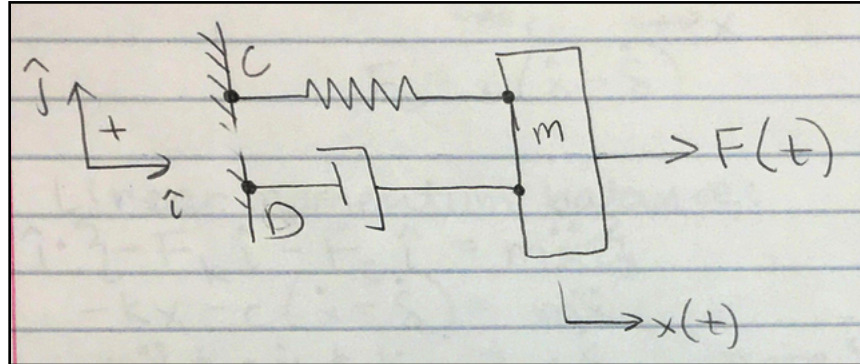


All Basic Facts of Damped, Forced, Harmonic Oscillators.

Problem 35: The harmonically forced vibration of a damped oscillator is given by this equation:

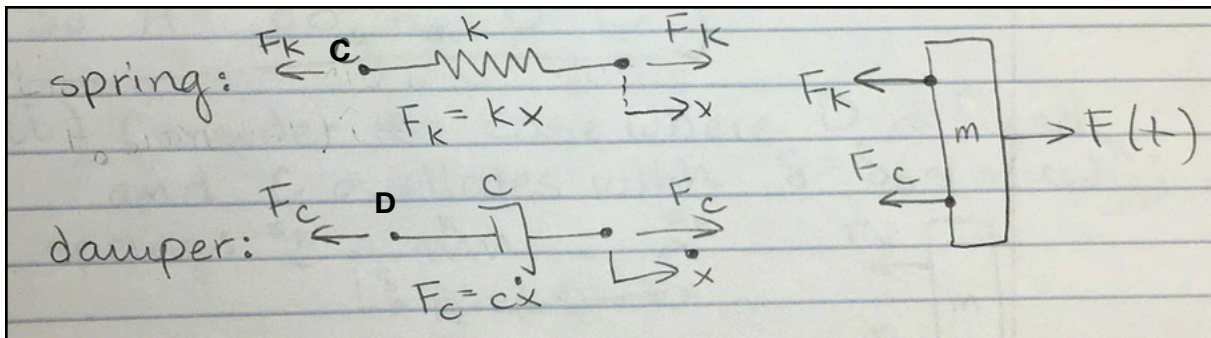
$$m\ddot{x} + c\dot{x} + kx = A \cos(\omega_0 t) + B \sin(\omega_0 t)$$

- a) Assume that the mass is connected to the spring and to the dashpot, the other ends of which are at C and D, respectively.



Define x as the displacement of the mass in inertial space. For each of the cases below, find A and B. The latter three cases are from excitation by a moving base.

- i) C and D are fixed and a force $F = F_0 \sin(\omega_0 t)$ acts on the mass. This scenario is illustrated in the sketch below.



Determining the equation of motion for the mass requires a linear momentum balance:

$$F(t)\hat{i} - F_k\hat{i} - F_c\hat{i} = m\ddot{x}\hat{i}$$

Dotting this equation with $\hat{i}$:

$$\hat{i} \cdot [F(t)\hat{i} - F_k\hat{i} - F_c\hat{i} = m\ddot{x}\hat{i}]$$

$$F(t) - F_k - F_c = m\ddot{x}$$

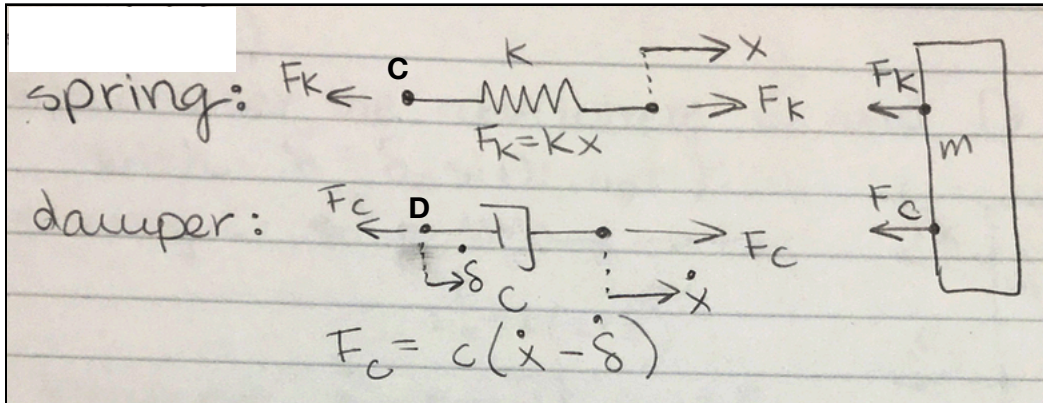
Inserting the forces illustrated in the previous sketch and rearranging the equation:

$$F_0 \sin(\omega_0 t) - kx - c\dot{x} = m\ddot{x}$$

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin(\omega_0 t)$$

Therefore, $A = 0, B = F_0$.

- ii) C is fixed and D oscillates with $\delta = \delta_0 \sin(\omega_0 t)$. This scenario is illustrated in the sketch below.



Determining the equation of motion for the mass requires a linear momentum balance:

$$-F_k \hat{i} - F_c \hat{i} = m \ddot{x} \hat{i}$$

Dotting this equation with $\hat{i}$:

$$\hat{i} \cdot [-F_k \hat{i} - F_c \hat{i} = m \ddot{x} \hat{i}]$$

$$-F_k - F_c = m \ddot{x}$$

Inserting the forces illustrated in the previous sketch and rearranging the equation:

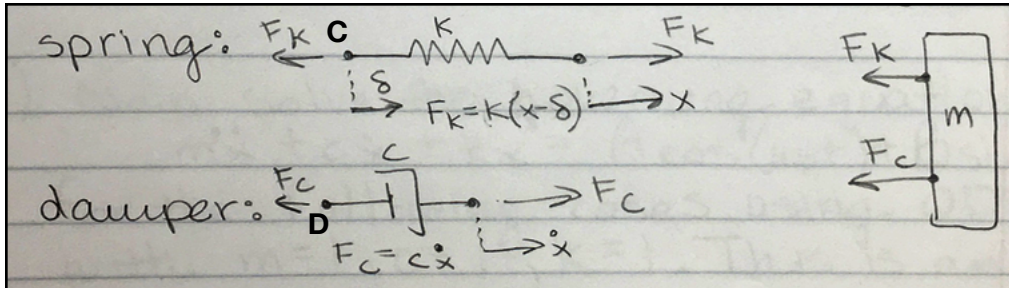
$$-kx - c(\dot{x} - \dot{\delta}) = m \ddot{x}$$

$$m \ddot{x} + c \dot{x} + kx = c \dot{\delta} \text{ where } \dot{\delta} = \delta_0 \omega_0 \cos(\omega_0 t)$$

$$m \ddot{x} + c \dot{x} + kx = c \delta_0 \omega_0 \cos(\omega_0 t)$$

$$\text{Therefore, } A = c \delta_0 \omega_0, B = 0.$$

- iii) D is fixed and C oscillates with $\delta = \delta_0 \sin(\omega_0 t)$. This scenario is illustrated in the sketch below:



Determining the equation of motion for the mass requires a linear momentum balance:

$$-F_k \hat{i} - F_c \hat{i} = m \ddot{x} \hat{i}$$

Dotting this equation with $\hat{i}$:

$$\hat{i} \cdot [-F_k \hat{i} - F_c \hat{i} = m \ddot{x} \hat{i}]$$

$$-F_k - F_c = m \ddot{x}$$

Inserting the forces illustrated in the previous sketch and rearranging the equation:

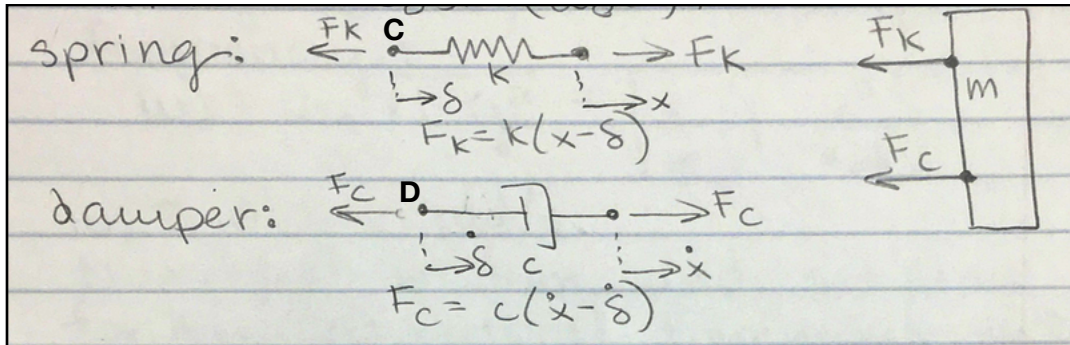
$$-k(x - \delta) - c\dot{x} = m\ddot{x}$$

$$m\ddot{x} + c\dot{x} + kx = k\delta \text{ where } \delta = \delta_0 \sin(\omega_0 t)$$

$$m\ddot{x} + c\dot{x} + kx = k\delta_0 \sin(\omega_0 t)$$

Therefore, $A = 0$, $B = k\delta_0$.

- iv) C and D oscillate together with $\delta = \delta_0 \sin(\omega_0 t)$. This scenario is illustrated in the sketch below.



Determining the equation of motion for the mass requires a linear momentum balance:

$$-F_k \hat{i} - F_c \hat{i} = m \ddot{x} \hat{i}$$

Dotting this equation with $\hat{i}$:

$$\hat{i} \cdot [-F_k \hat{i} - F_c \hat{i} = m \ddot{x} \hat{i}]$$

$$-F_k - F_c = m \ddot{x}$$

Inserting the forces illustrated in the previous sketch and rearranging the equation:

$$-k(x - \delta) - c(\dot{x} - \dot{\delta}) = m \ddot{x}$$

$$m \ddot{x} + c \dot{x} + kx = c \dot{\delta} + k\delta \text{ where } \delta = \delta_0 \sin(\omega_0 t) \text{ and } \dot{\delta} = \delta_0 \omega_0 \cos(\omega_0 t)$$

$$m \ddot{x} + c \dot{x} + kx = c \delta_0 \omega_0 \cos(\omega_0 t) + k \delta_0 \sin(\omega_0 t)$$

$$\text{Therefore, } A = c \delta_0 \omega_0, B = k \delta_0.$$

- b) For the following problems, solve the governing equation above numerically using, say ODE45 using various appropriate forcing and initial conditions. For definiteness use the underdamped case $m = 1, c = 1, k = 1$. **(All of the MATLAB scripts and functions used for part (b) are labeled and attached at the end of this document.)**
- i) Set $A = 0, B = 0$, and $c = 0$. Using numerics find the natural frequency ω_n . Do this using 'events' (for the mass released from rest, find the time until the velocity gets to zero from above). Then calculate the frequency from this measure period. Compare the result with the analytic result $\omega_n = \sqrt{k/m}$.

With the parameters listed above, the equation of motion for this mass is:

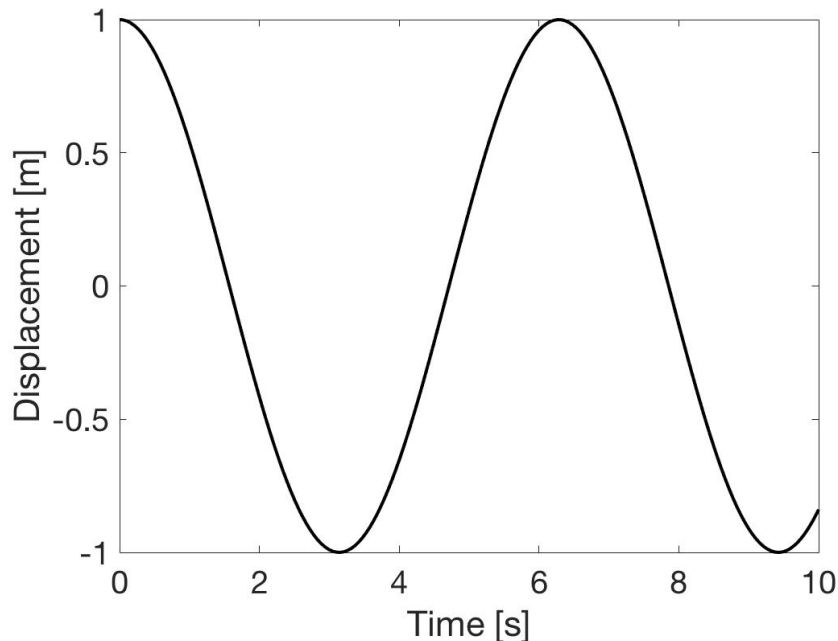
$$m\ddot{x} + kx = 0$$

The analytical solution for the natural frequency is

$$\omega_{n,a} = \sqrt{\frac{k}{m}} = 1 \frac{rad}{s}$$

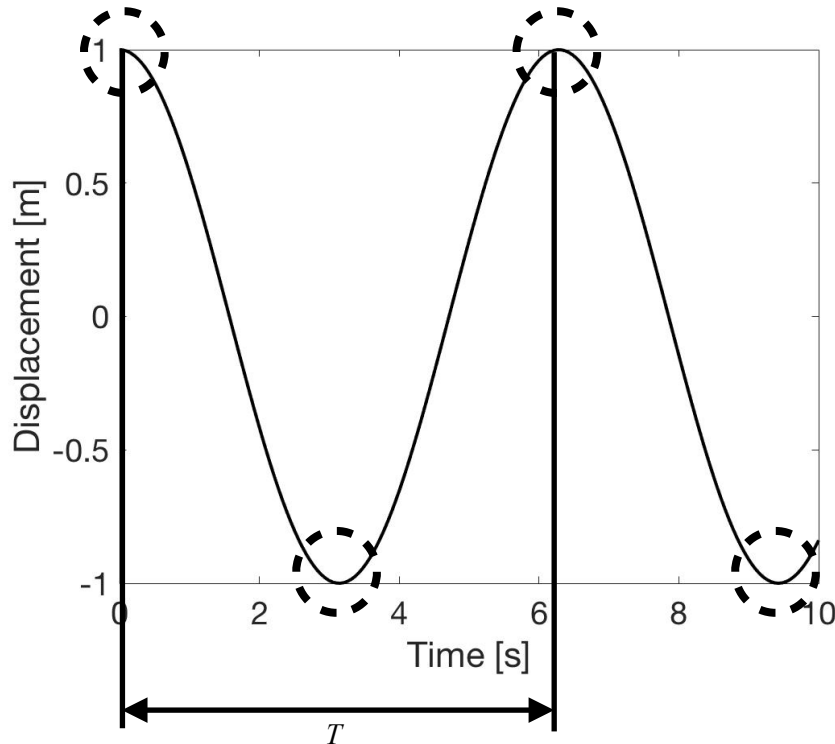
The numerical solution with the following initial conditions is shown below.

$$x(0) = 1 \text{ m}, \dot{x}(0) = 0 \frac{m}{s}, m = 1 \text{ kg}, k = 1 \frac{N}{m}$$



Using the 'events' option with ODE45, the undamped natural frequency of this solution can be determined. The 'events' function is an option with ODE45 that can detect when certain events occur during a numerical solution, particularly when certain values are equal to zero.

In this case, the events we are looking for are the times where the velocity of the mass is equal to zero. These locations are circled in the plot on the following page.



In order to calculate the undamped natural frequency with these numerical results, we need the period of the motion, T , labeled in the plot above. Using ‘events’ to detect the times where the velocity is equal to zero allows us to find the period of the motion.

For this scenario, MATLAB recorded the times listed in the following vector `te` as the times where $\dot{x}(t)$ is equal to zero.

`te =`

```
0.0000000000000000
3.141592286999844
6.283184994081370
9.424777433225220
```

You can see that the times listed in `te` agree with the plot labeled above where the velocity of the mass is equal to zero.

The entry in `te` that agrees with the period labeled in the plot above is 6.283184994081370 [s]. Finally, the numerical solution for the undamped natural frequency is:

$$\omega_{n,n} = \frac{2\pi}{T} = \frac{2\pi}{6.283184994081370} = 1.000000049831131 \frac{rad}{s}$$

As a reminder, the analytical solution for the undamped natural frequency is:

$$\omega_{n,n} = \sqrt{\frac{k}{m}} = 1 \frac{rad}{s}$$

The difference between the numerical and analytical solutions for undamped natural frequency is:

$$difference = 1.000000049831131 \frac{rad}{s} - 1 \frac{rad}{s} = 4.983113122136729 \times 10^{-8} \frac{rad}{s}$$

The absolute and relative tolerances used in the numerical solution with ODE45 were both 1×10^{-6} . The difference between the numerical and analytical solutions for the undamped natural frequency is within this numerical tolerance.

- ii) Now set $A = 0$, $B = 0$, and $c = 1$ and find the damped natural frequency ω_d . Compare this with the analytic result.

With the parameters listed above, the equation of motion for this mass is:

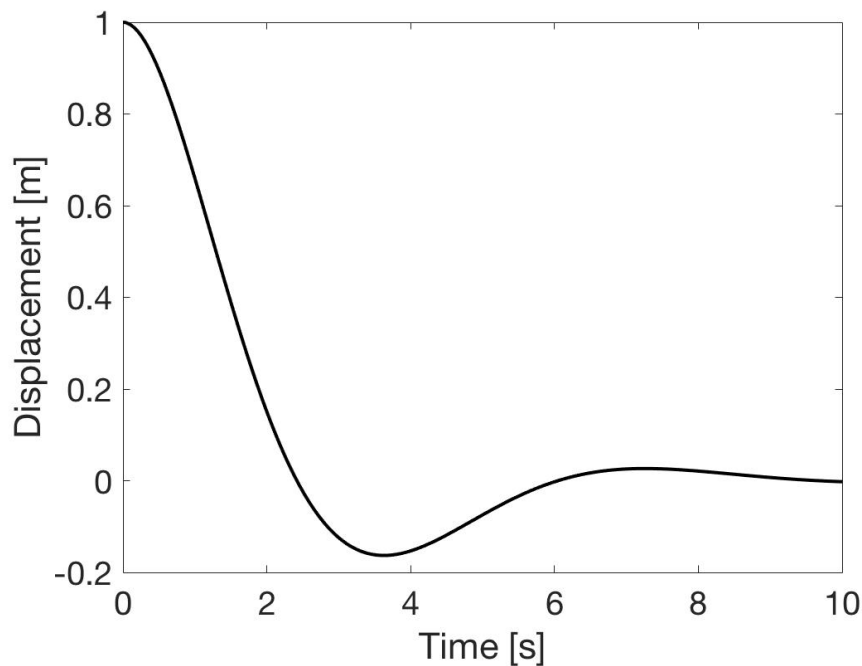
$$m\ddot{x} + c\dot{x} + kx = 0$$

The analytical solution for the damped natural frequency is

$$\omega_{d,a} = \omega_n \sqrt{1 - \zeta^2} = \sqrt{\frac{k}{m} \left(1 - \frac{c^2}{4mk}\right)} = 0.866025403784439 \frac{rad}{s}$$

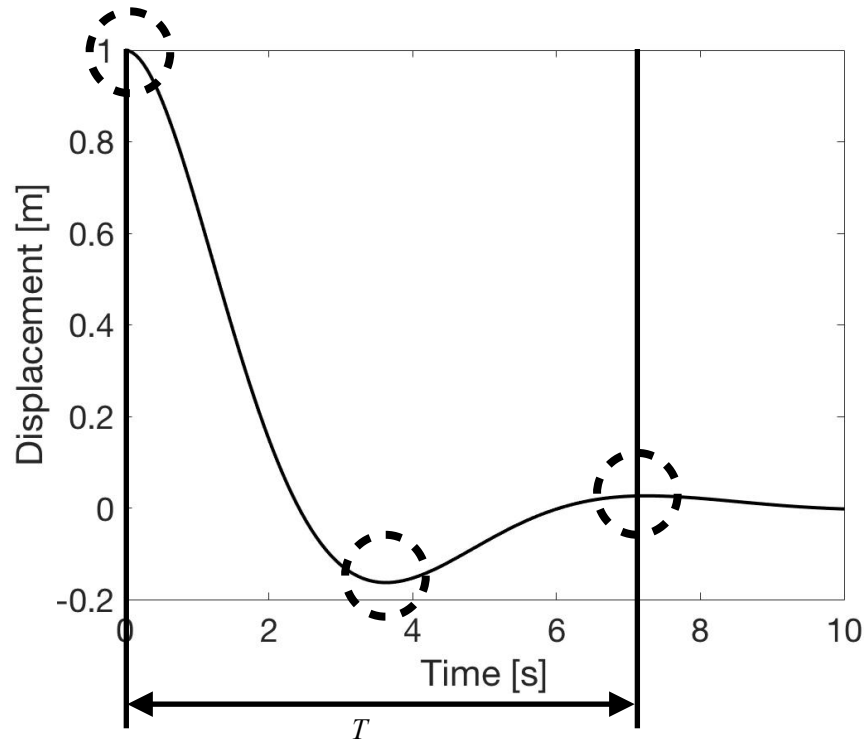
The numerical solution for displacement is shown below with the following initial conditions.

$$x(0) = 1 \text{ m}, \dot{x}(0) = 0 \frac{m}{s}, m = 1 \text{ kg}, c = 1 \frac{Ns}{m}, k = 1 \frac{N}{m}$$



Using the 'events' option with ODE45, the damped natural frequency of this solution can be determined. Recall that the 'events' function is an option with ODE45 that can detect when certain events occur during a numerical solution, particularly when certain values are equal to zero.

Again, in this case, the events we are looking for are the times where the velocity of the mass is equal to zero. These locations are circled in the plot on the following page.



In order to calculate the damped natural frequency with these numerical results, we need the period of the motion, T , labeled in the plot above. Using 'events' to detect the times where the velocity is equal to zero allows us to find the period of the motion.

For this scenario, MATLAB recorded the times listed in the following vector `te` as the times where $\dot{x}(t)$ is equal to zero.

`te =`

```
0.0000000000000000
3.627599327525224
7.255205398330033
```

You can see that the times listed in `te` agree with the plot labeled above where the velocity of the mass is equal to zero.

The entry in `te` that agrees with the period labeled in the plot above is 7.255205398330033 [s]. Finally, the numerical solution for the undamped natural frequency is:

$$\omega_{d,n} = \frac{2\pi}{T} = \frac{2\pi}{7.255205398330033} = 0.866024455851494 \frac{rad}{s}$$

As a reminder, the analytical solution for the damped natural frequency is:

$$\omega_{d,a} = \omega_n \sqrt{1 - \zeta^2} = \sqrt{\frac{k}{m} \left(1 - \frac{c^2}{4mk}\right)} = 0.866025403784439 \frac{rad}{s}$$

The difference between the numerical and analytical solutions for damped natural frequency is:

$$\begin{aligned} difference &= 0.866024455851494 \frac{rad}{s} - 0.866025403784439 \frac{rad}{s} \\ &= -9.479329449169072 \times 10^{-7} \frac{rad}{s} \end{aligned}$$

The absolute and relative tolerances used in the numerical solution with ODE45 were both 1×10^{-6} . The difference between the numerical and analytical solutions for the undamped natural frequency is within this numerical tolerance.

- iii) Using 'events' that do not terminate the integration, and using the logarithmic decrement method, find the damping ratio ζ . Compare with the analytic result.

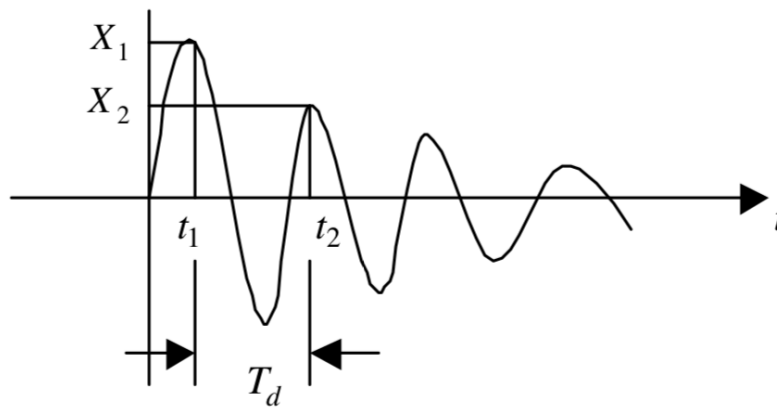
Recall the equation of motion for this damped harmonic system:

$$m\ddot{x} + c\dot{x} + kx = 0$$

The analytical solution for the damped natural frequency is

$$\zeta_a = \frac{c}{\sqrt{4mk}} = 0.5000000000000000$$

The damping ratio can also be found numerically by using the logarithmic decrement method. This method consists of first collecting the amplitudes of the first two peaks in a damped harmonic motion:



Once you have collected these values, the logarithmic decrement is calculated as follows:

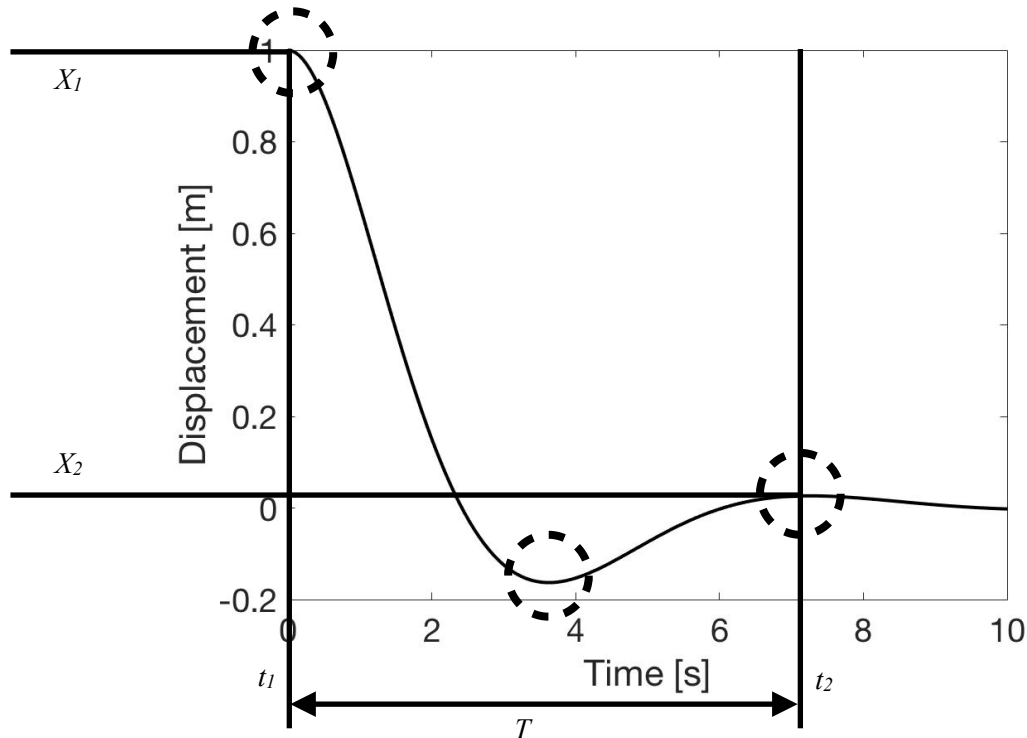
$$\delta = \ln \frac{X_1}{X_2}$$

Finally, the damping ratio is calculated using the logarithmic decrement value previously calculated:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

The 'events' option in MATLAB can be applied again here to determine the first two amplitudes of this motion by detecting the times that the mass is at a velocity of zero.

The plot on the following page illustrates the amplitudes and times collected for the logarithmic decrement method in this case as well as the times that the mass is at zero velocity.



Recall that in order to calculate the logarithmic decrement and, as a result, the damping ratio, we need X_1 and X_2 .

MATLAB recorded the times where $\dot{x}(t)$ was equal to zero and the values of $x(t)$ and $\dot{x}(t)$ at these times are listed below with $x(t)$ in the left column and $\dot{x}(t)$ in the right column.

ye =

1.0000000000000000	-0.0000000000000000
-0.163033638899176	0.0000000000000000
0.026579885640098	-0.0000000000000002

You can see that the displacement values in the left column agree with the plot labeled above where the velocity of the mass is equal to zero.

The entries in the first column that correspond with X_1 and X_2 are 1.0000000000000000 [m] and 0.026579885640098 [m], respectively. As such, the logarithmic decrement is calculated as follows:

$$\delta = \ln \frac{X_1}{X_2} = \ln \frac{1.0000000000000000}{0.026579885640098} = 3.627600528185661$$

Finally, the numerical solution for the damping ratio is:

$$\zeta_n = \frac{3.627600528185661}{\sqrt{4\pi^2 + (3.627600528185661)^2}} = 0.500000186044236$$

$$\zeta_a = \frac{c}{\sqrt{4mk}} = 0.5000000000000000$$

The difference between the numerical and analytical solutions for the damping ratio is:

$$\begin{aligned} \text{difference} &= 0.500000186044236 - 0.5000000000000000 \\ &= 1.860442360168690 \times 10^{-7} \end{aligned}$$

The absolute and relative tolerances used in the numerical solution with ODE45 were both 1×10^{-6} . The difference between the numerical and analytical solutions for the damping ratio is within this numerical tolerance.

- iv) Draw a frequency response curve (each point on this curve requires a full simulation). For example, use A or $B = 0$ and the other equal to 1 and look at the amplitude of steady state response. The hard part here is running the simulation long enough so that the response is “steady state”. Compare this curve with an analytically derived curve. Using the numerics, find as accurately as you can, the frequency at which the amplitude of the steady state response is maximum.

The following equation of motion can be used to create a frequency response curve for this part:

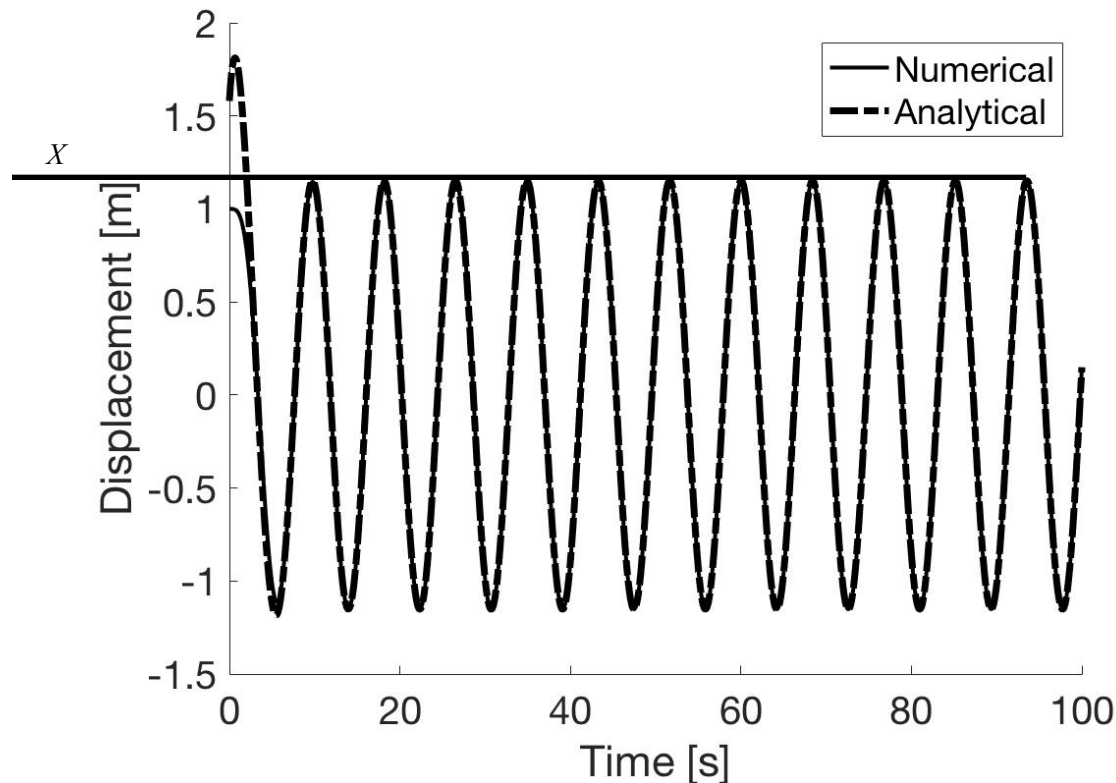
$$m\ddot{x} + c\dot{x} + kx = \cos(\omega_0 t)$$

The analytic solution for this equation of motion is:

$$x(t) = e^{-\zeta\omega_n t} \left\{ x_0 \cos(\omega_d t) + \frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d} \sin(\omega_d t) \right\} + X \cos(\omega_0 t - \phi)$$

$$\text{where } X = \frac{1}{\sqrt{(k - m\omega_0^2)^2 + c^2\omega_0^2}} \text{ and } \phi = \tan^{-1} \frac{c\omega_0}{k - m\omega_0^2}$$

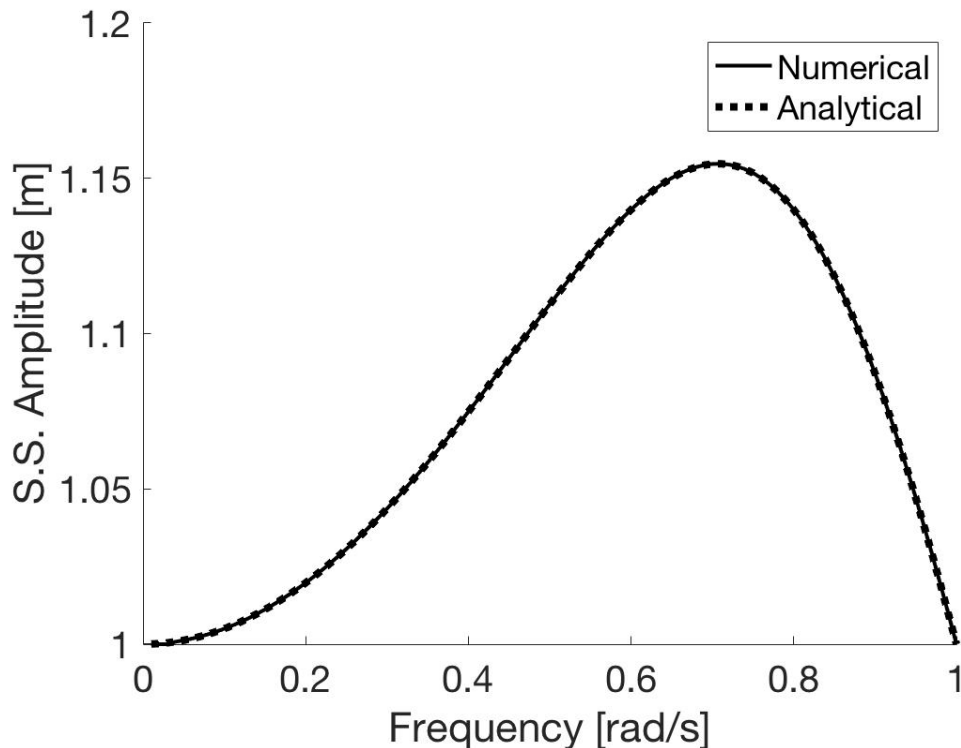
For example, you can see below the numerical and analytical steady state behavior for this system under the application of an excitation frequency ω_0 of 0.75 [rad/s].



The steady state amplitude is the amplitude of the motion labeled as X .

We are looking for the resonant frequency for this system, the excitation frequency that will produce the largest steady state amplitude. In order to determine this excitation frequency, a frequency response curve must be created and analyzed to find the excitation frequency at which the maximum steady state amplitude occurs.

The steady state amplitude for both the numerical solution and the analytical solution can be determined with MATLAB for all excitation frequencies between 0 [rad/s] and 1 [rad/s]. Below is the excitation frequency response for both the numerical and analytical solutions for excitation frequencies between 0 [rad/s] and 1 [rad/s].



For the numerical solution, the maximum steady state amplitude is 1.154687388025473 [m] and occurs at an excitation frequency of 0.710000000000000 [rad/s].

For the analytical solution, the maximum steady state amplitude is 1.154687598252741 [m] and occurs at an excitation frequency of 0.710000000000000 [rad/s].

- v) Compare the three frequencies: 1) natural frequency, 2) damped natural frequency, and 3) 'resonance' frequency (frequency which gives maximum amplitude response). Note their order and note how close, or not, they are to each other.

The undamped natural frequency, damped natural frequency, and resonant frequency determined for this system with a numerical solution are listed below.

$$\omega_n = 1.000000049831131 \frac{rad}{s}$$

$$\omega_d = 0.866024455851494 \frac{rad}{s}$$

$$\omega_r = 0.7100000000000000 \frac{rad}{s}$$

As seen above, $\omega_n > \omega_d > \omega_r$.

These values are fairly close. Quantitatively, ω_n is only 0.134 [rad/s] larger than ω_d and ω_d is only 0.156 [rad/s] larger than ω_r .


```
% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

clear all
close all
clc

% fixed constants
m = 1; % [kg]
c = 1; % [N-s/m]
k = 1; % [N/m]

% initial conditions
xi = 1;
xdoti = 0;
zi = [xi xdoti];

% calling ODE45 for numerical solution
tEnd = 10;
tspan = [0 tEnd];
options = odeset('reltol',1e-6,'abstol',1e-6,'Events',@EventsFcnI);
[t z te ye ie] = ode45(@(tSol,zSol) RHSfxBI(tSol,zSol,m,c,k),tspan,zi,options);
x = z(:,1);
xdot = z(:,2);
te % printing times of zero velocity to command window

% plotting numerical solution for displacement
figure
plot(t,x,'k','LineWidth',2)
set(gca,'FontSize',20)
xlabel('Time [s]')
ylabel('Displacement [m]')
```

```
function [result position isterminal direction] = RHSfxBI(t,z,m,c,k)
% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

x = z(1); % unpacking variables
xdot = z(2);
xddot = -k*x/m;

result = zeros(2,1);
result(1) = xdot;
result(2) = xddot;

[value,isterminal,direction] = EventsFcnII(t,z); % calling 'events'
% functions to find times of zero velocity

end
```

```
function [position,isterminal,direction] = EventsFcnI(t,y)
    % Maura O'Neill
    % MAE 5730
    % HW 11
    % Problem 35

    position = y(2); % The value that we want to be zero
    isterminal = 0; % Halt integration
    direction = 0; % The zero can be approached from either direction
end
```

```
% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

clear all
close all
clc

% fixed constants
m = 1; % [kg]
c = 1; % [N-s/m]
k = 1; % [N/m]

% initial conditions
xi = 1;
xdoti = 0;
zi = [xi xdoti];

% numerical solution
tEnd = 10;
tspan = [0 tEnd];
options = odeset('reltol',1e-6,'abstol',1e-6,'Events',@EventsFcnII);
[t z te ye ie] = ode45(@(tSol,zSol) RHSfxBII(tSol,zSol,m,c,k),tspan,zi,options);
x = z(:,1);
xdot = z(:,2);
te % printing times of zero velocity to command window

% plotting numerical solution for displacement
figure
plot(t,x,'k','LineWidth',2)
set(gca,'FontSize',20)
xlabel('Time [s]')
ylabel('Displacement [m]')
```

```
function [result position isterminal direction] = RHSfxBII(t,z,m,c,k)
% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

x = z(1); % unpacking variables
xdot = z(2);
xddot = (-k*x-c*xdot)/m;

result = zeros(2,1);
result(1) = xdot;
result(2) = xddot;

% calling 'events' function to determine times of zero velocity
[value,isterminal,direction] = EventsFcnII(t,z);

end
```

```
function [position, isterminal, direction] = EventsFcnII(t,y)
    % Maura O'Neill
    % MAE 5730
    % HW 11
    % Problem 35

    position = y(2); % The value that we want to be zero
    isterminal = 0; % Halt integration
    direction = 0; % The zero can be approached from either direction
end
```

```
% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

clear all
close all
clc

% fixed constants
m = 1; % [kg]
c = 1; % [N-s/m]
k = 1; % [N/m]

% initial conditions
xi = 1;
xdoti = 0;
zi = [xi xdoti];

% numerical solution
tEnd = 10;
tspan = [0 tEnd];
options = odeset('reltol',1e-6,'abstol',1e-6,'Events',@EventsFcnIII);
[t z te ye ie] = ode45(@(tSol,zSol) RHSfxBIII(tSol,zSol,m,c,k),tspan,zi, options);
x = z(:,1);
xdot = z(:,2);
te % printing times of zero velocity to command window
ye % printing displacement and velocity values at times of zero velocity
% to command window

% calculating logarithmic decrement and damping ratio
% as well as printing these values to the command window
dec = log(ye(1)/ye(3))
ratio = dec/sqrt(4*pi^2+dec^2)

% plotting numerical solution for displacement
figure
plot(t,x,'k','LineWidth',2)
set(gca,'FontSize',20)
xlabel('Time [s]')
ylabel('Displacement [m]')
```

```
function [result position isterminal direction] = RHSfxBIII(t,z,m,c,k)
% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

x = z(1); % unpacking variables
xdot = z(2);
xddot = (-k*x-c*xdot)/m;

result = zeros(2,1);
result(1) = xdot;
result(2) = xddot;

% calling 'events' function to determine times of zero velocity
[value,isterminal,direction] = EventsFcnIII(t,z);

end
```



```
function [position,isterminal,direction] = EventsFcnIII(t,y)
    % Maura O'Neill
    % MAE 5730
    % HW 11
    % Problem 35

    position = y(2); % The value that we want to be zero
    isterminal = 0; % Halt integration
    direction = 0; % The zero can be approached from either direction
end
```

```

% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

clear all
close all
clc

% fixed constants
m = 1; % [kg]
c = 1; % [N-s/m]
k = 1; % [N/m]

% initial conditions
xi = 1;
xdoti = 0;
zi = [xi xdoti];

% creating array of excitation frequency values to create
% frequency response curve
w0 = 0.01:0.01:1;

% numerical and analytical solutions for different values of
% excitation frequencies
for i = 1:length(w0)
    w0curr = w0(i);
    tEnd = 100;
    tspan = [0 tEnd];
    options = odeset('reltol',1e-6,'abstol',1e-6,'Events',@EventsFcnIV);
    [t z te ye ie] = ode45(@(tSol,zSol) RHSfxBIV(tSol,zSol,m,c,k,w0curr),...
        tspan,zi,options);
    x = z(:,1);
    xdot = z(:,2);
    D = size(ye);
    ssAmp(i) = abs(ye(D(1),1)); % looking for steady state amplitude value
    % of numerical solution using results of 'events' function

    % calculating analytical damping ratio, natural frequency, and
    % damped natural frequency
    ratio = c/sqrt(4*m*k);
    wn = sqrt(k/m);
    wd = wn*sqrt(1-ratio^2);

    % calculating phi and X of particular solution at each value
    % for excitation frequency
    phi(i) = atan(c*w0(i)/(k-m*w0(i)^2));
    X(i) = 1/sqrt((k-m*w0(i)^2)^2+c^2*w0(i)^2);
end

% plotting excitation frequency response for both numerical and analytical
% solutions
figure
hold on
plot(w0,ssAmp,'k','LineWidth',2)
plot(w0,X,'k:','LineWidth',4)
set(gca,'FontSize',20)
xlabel('Frequency [rad/s]')
ylabel('S.S. Amplitude [m]')
legend('Numerical','Analytical')

% determining maximum steady state amplitude from numerical solution
% for excitation frequency response and printing this value to the

```

```
% command window  
[maxAmp, indAmp] = max(ssAmp)  
maxFreq = w0(indAmp)
```

```
function [result position isterminal direction] = RHSfxBIVfr(t,z,m,c,k,w0)
% Maura O'Neill
% MAE 5730
% HW 11
% Problem 35

x = z(1); % unpacking variables
xdot = z(2);
xddot = (-k*x-c*xdot+cos(w0*t))/m;

result = zeros(2,1);
result(1) = xdot;
result(2) = xddot;

% calling 'events' function for determining times of zero velocity
[value,isterminal,direction] = EventsFcnIV(t,z);

end
```

```
function [position,isterminal,direction] = EventsFcnIVfr(t,y)
    % Maura O'Neill
    % MAE 5730
    % HW 11
    % Problem 35

    position = y(2); % The value that we want to be zero
    isterminal = 0; % Halt integration
    direction = 0; % The zero can be approached from either direction
end
```